

Broccoli

CULTIVARS

- Batavia
- Marathon
- Imperial (summer)

INTENSIVE SPACING

- 2 rows per bed spaced every 18" (45 cm) on the row

FERTILIZATION

- Broccoli, cabbage and cauliflowers are all heavy feeders that require lots of compost, +/- 1 inch (5 cm) all across the bed. In rich soils, compost can be replaced with a buried cover crop of leguminous nature, but in both cases, we add nitrogen-rich organic fertilizers like pelleted chicken manure (5-3-2) to help the plant quickly develop its canopy.
- A second boost of nitrogen is added to the crop +/- 14 days after the transplant by amending the soil with blood or alfalfa meal.

IRRIGATION

Irrigation Type

- Drip tape or sprinkler

Irrigation Management

- Root depth: Shallow (6"-24")
- Critical period: Head development.
- General: Commercial growers use 1-1-1/2" per week. Extra water during crown will add bulk to the harvest.

IMPLANTATION

For weekly supply throughout the growing season, we plant 2 different cultivars every 15 days. Each of these cultivars has a different DTM (days to maturity) which stagger the harvest window. We plant them with other heavy feeding crucifers like cauliflowers, cabbages and kohlrabi, all crops which spend +/- 60 days in the field.

For summer plantings, plan for heat tolerant cultivars.

EQUIPMENT

- Rotary harrow
- Broadfork
- 30" bed prep rake + marking tubes
- Declumping tray
- Anti-insect net and hoops

ITINERARY

- Remove tarp and-or clean large debris. Use the rake if needed. *
- Uniformly apply compost and amendments and pass with the rotary harrow at a depth of 2" (5 cm). To make transplanting easier (seeing as the soil is loose after being harrowed), we only prepare the beds when we are about to transplant.
- Using the bed prep rake and 2 tubes, mark the rows both lengthways and sideways. Bring the well-watered trays.
- Using the declumping tray to facilitate clump extraction, 1 person will drop the seedling while 2 others transplants the plants as fast as possible. 3 people/100'bed is good ratio for efficient workflow.
- Bury the seedling all the way to where the leaves start to branch out.
- Install the hoops every 5' (1.5 m) and cover the crops with the anti-insect net. Set sand bags on every hoop.
- Water the bed thoroughly the first time, keeping the soil moist for the first week after the transplant.

* If the previous crop is too well established, or if the bed is too messy, mow the crop as low as possible and harrow at a depth of 2" (5 cm). Add 1" (2.5 cm) of compost evenly spread on the whole width of the bed and transplant into compost following a no-till strategy.

WEEDING

- 14 days after the transplant, the crops are cultivated using a stirrup hoe and amended bed with blood or alfalfa meal. Hoeing will take care of the weeds, break the crust and mix your amendments in the soil.
- Cover the beds with a +/-4" of straw mulch (10 cm). This will keep the soil cooler, which is beneficial for the crop.

PLANT PROTECTION

- Immediately after transplanting, install anti-insect net on the crop. Place sandbags every 5' (1.5 M) to make sure the nets are sealed down.
- After transplant, seedlings are sometimes being attacked by white or gray worms in which case, they'll need to be replaced. For this reason, keep leftovers seedlings another week in the nursery after the transplant.
- Slugs can also be an issue. But, unless there's a major infestation, we don't worry about them.
- To protect the crop against the cabbage moth, systematically spray BT +/- 25 days and another 10 days later. When spraying for BT, add boron and seaweed (or compost tea) together and apply foliarly. This is better done early morning so to let the leaf absorb the foliar application. Avoid spraying during or before a rain, as it will diminish the effect of the application.
- If sweet midge is a problem or if flea beetles are very problematic, crops should be covered with insect netting all the way to harvest.

HARVEST

- Broccoli is harvested standing upright using a machete field knife and tree planting bags. Once the first florets ripe, harvest should be done every day (or other day).
- Cut the plants with 3 inch of stem, leaving 2 or 3 small leaves on the plant.

- You know when the broccoli is ready not by its size, but rather by the appearance of its budding seeds. Florets should be a dark green and firm. Don't harvest over matured heads or those with traces of head rot.
- The harvest is brought immediately back to the washing station when completed.

CONDITIONING

- Dump broccolis in cold water, let drain for a minute and store in storage bins.
- Identify the bin with the date of the harvest.

STORING

- Broccolis are stored in cold room keep at 35 °F (2 °C) and should not be stored for more than 4 days before being sold.